

Label-Efficient Dark-Vessel Detection in SAR: Does SAR-Domain Pretraining Outperform Optical and ImageNet Transfer Across ViT and CNN?

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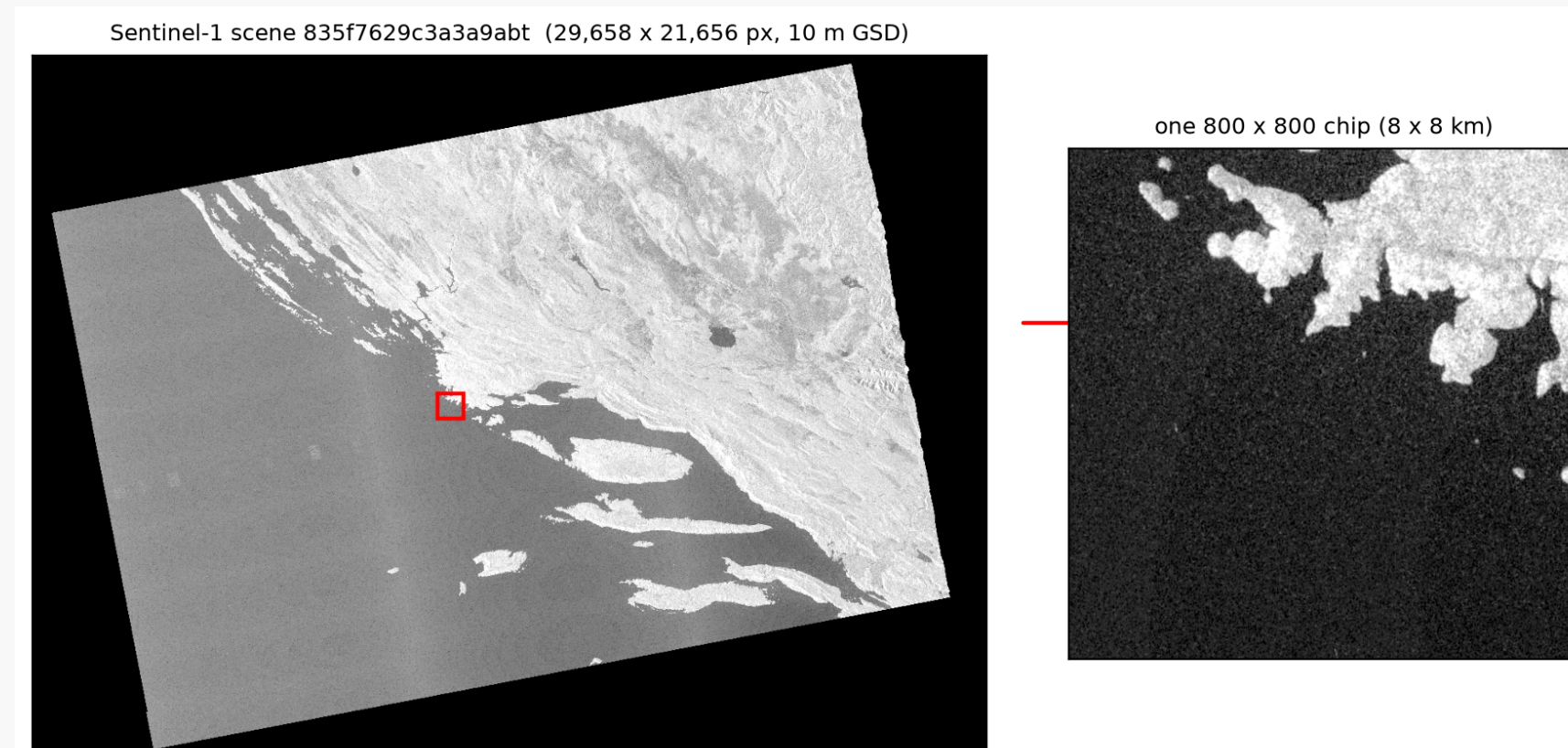
Why it matters

- **Dark vessels** have no matching AIS report. Sentinel-1 SAR sees them at night and through cloud.
- Vessels fill a few pixels in a 25,000-pixel scene. Sea clutter and coastlines give strong false returns.
- **Human-verified labels are scarce.** Pretraining can lower the label cost, but which source domain helps most?

Question

Does **SAR** pretraining beat **optical remote-sensing** and **ImageNet** pretraining when labels are few? Does the answer hold for both a ViT and a CNN?

Data: xView3-SAR



One Sentinel-1 scene (about 300 × 220 km) and one 800-pixel chip (8 × 8 km).

- 150 frozen study scenes: **111 train, 23 dev, 16 test**.
- Nested budgets: **12 < 28 < 56 < 111** training scenes (10, 25, 50, 100%).
- **50 near-shore, human-verified scenes**, opened once after test scoring: 8,642 vessels, 3,644 dark, 2,686 within 2 km of shore.
- Input [VH, VV, VH - VV] in dB, 10 m pixels.

Fix before the final cohort

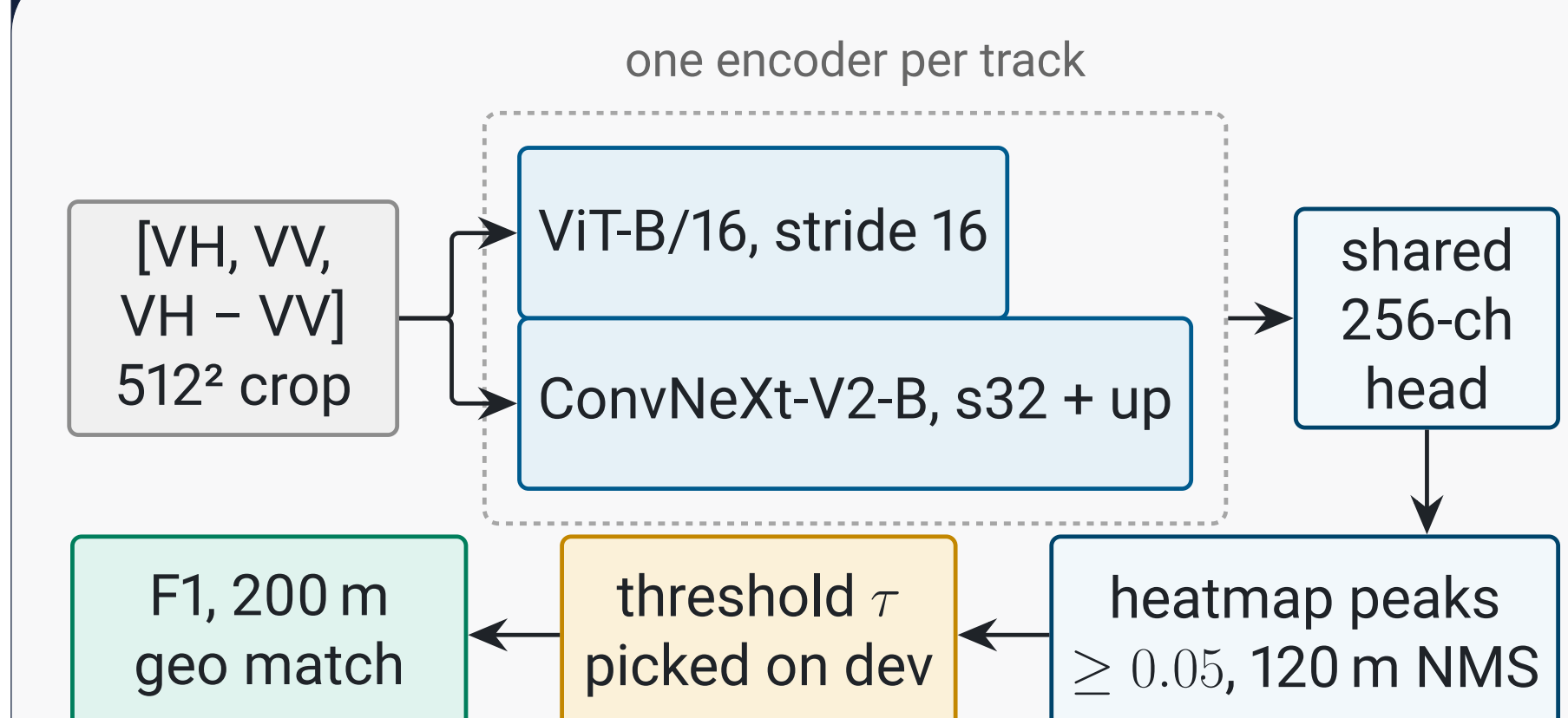
BigEarthNet-S2 expects 10 optical bands. The first adapter kept 3 band kernels and scaled them by 10/3: first-layer weight norm **5.85×**. A seeded 3-channel stem plus every other transferred tensor raised dev F1 from 0.799 to 0.837 (12 scenes). We then retrained **all 32 cells** from initialization.

8 arms × 4 budgets = 32 cells

Role	ViT-B/16 (86M)	ConvNeXt-V2-B (89M)
Random	fresh init	fresh init
Optical	SatDINO (fMoW-RGB)	BigEarthNet-S2
SAR	SARMAE (SAR-1M)	BigEarthNet-S1
ImageNet	AugReg	FCMAE

Only the initialization changes inside a track. We compare domains **within** an architecture, then check whether the pattern repeats across the two.

One shared point detector

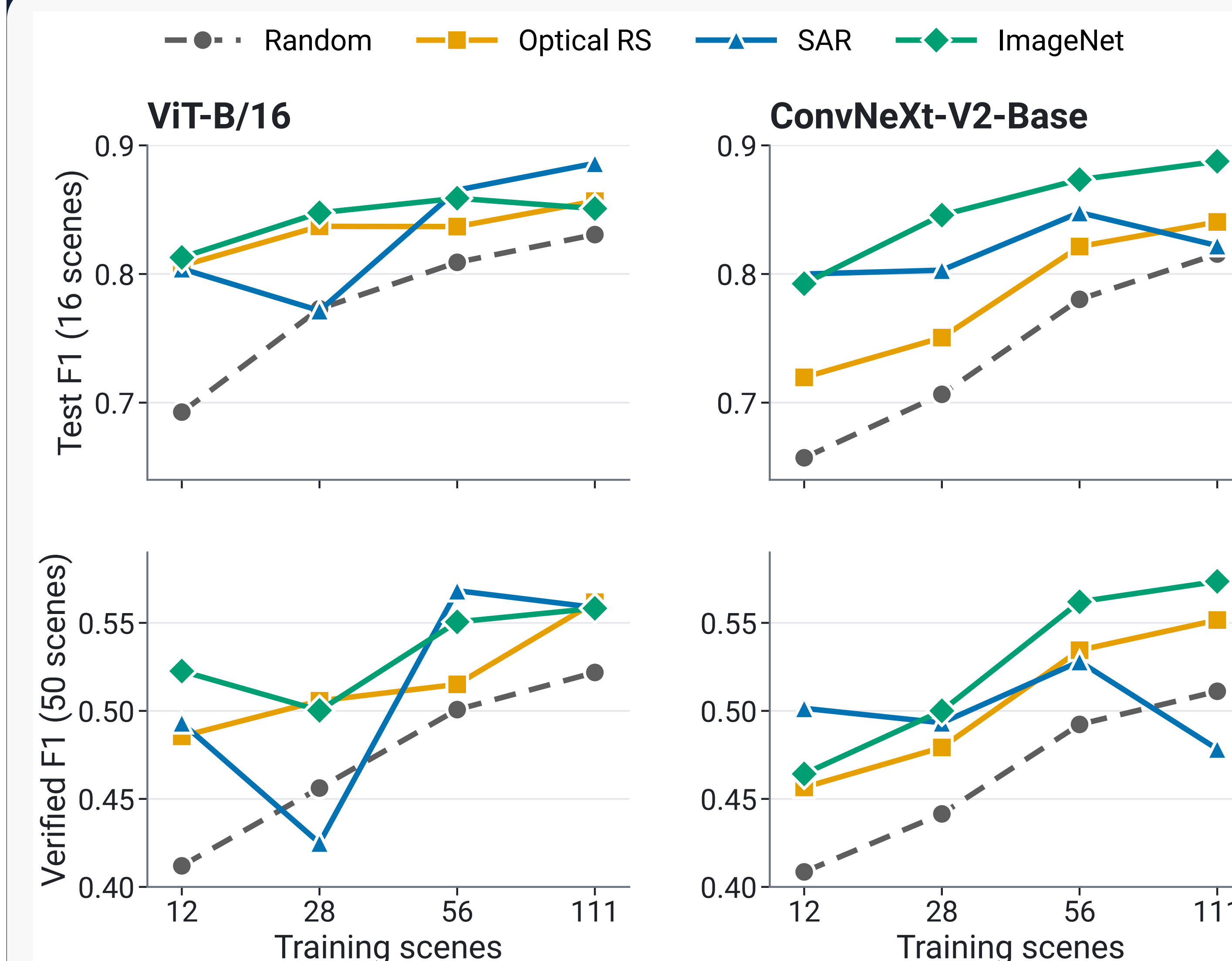


- CenterNet-style Gaussian heatmap and focal loss; F1 with 200 m geographic matching.
- Same optimizer, schedule, crops, data order and seed (0) for every cell. Strict FP32: 567.6 H100 GPU-hours.

Once-only held-out protocol

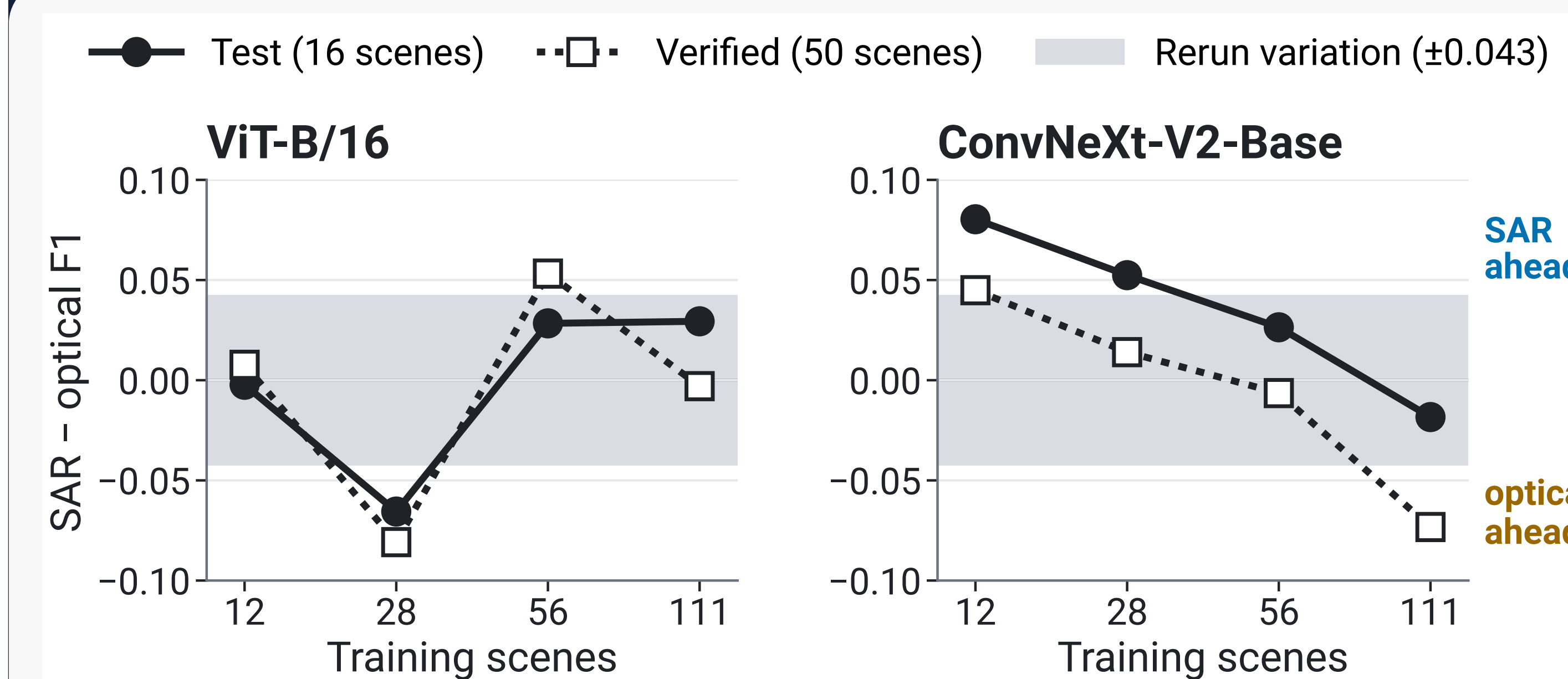
- 1 **Train 32 cells.** A sweep on 8 dev scenes picks each checkpoint and threshold.
- 2 **Freeze the cohort.** Hash-bind every marker and checkpoint.
- 3 **Score 16 test scenes once.** Same thresholds, no retuning.
- 4 **Open 50 verified scenes once.** Same thresholds, all 32 cells.

Pretraining helps most when labels are scarce



F1 with 200 m matching. Each of the 32 cells is scored once with the threshold it picked on 8 dev scenes. Top: 16 held-out test scenes. Bottom: 50 near-shore, human-verified scenes. Seed 0; no intervals.

SAR vs. optical depends on backbone and budget



Gray band: the largest test-F1 change across 28 cells that we retrained with the same recipe and seed (±0.043). A contrast inside the band is within rerun variation.

The verified scenes are a coastal stress test

0.409–0.574 ≤ **0.228** ≤ **0.041**

verified F1, all 32 cells
(test: 0.657–0.888)

dark-vessel recall
(3,644 dark vessels)

near-shore F1
(2,686 vessels within 2 km)

Precision stays at 0.639–0.909 while recall falls to 0.286–0.434: the detectors miss coastal vessels rather than over-fire.

Findings

12 ≈ 111

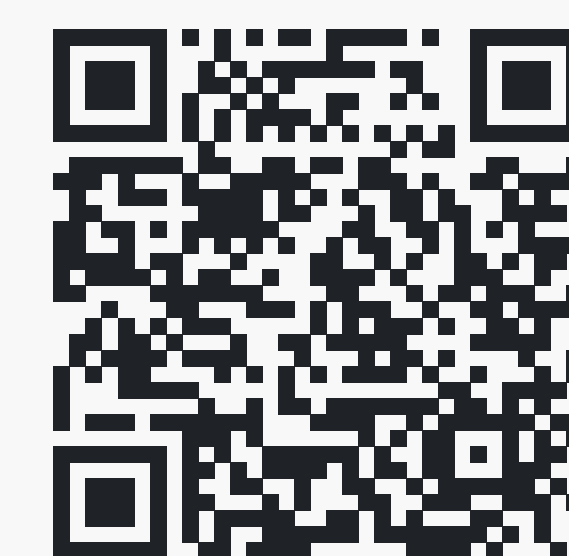
On the verified scenes, ImageNet ViT with 12 training scenes matches random ViT with 111 (0.523 vs. 0.522 F1).

- 1 **Pretraining pays off with few labels.** At 12 scenes every pretrained arm beats random: +0.063 to +0.143 test F1, +0.048 to +0.111 verified F1.
- 2 **SAR beats optical only while labels are scarce.** CNN: +0.080 test F1 at 12 scenes, −0.018 at 111. ViT: no stable order.
- 3 **The input adapter matters.** With a seeded 3-channel stem, BigEarthNet-S2 beats random at every budget (+0.025 to +0.063 test F1).
- 4 **Coastal and dark vessels stay hard.** Verified F1 is 0.409–0.574 (test: 0.657–0.888). Dark-vessel recall ≤ 0.228; near-shore F1 ≤ 0.041.

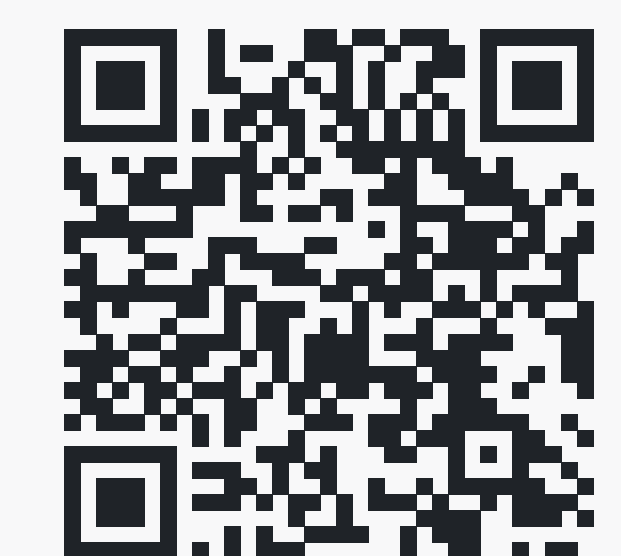
Limits

- One seed per cell; 8 dev scenes pick checkpoints and thresholds.
- The predeclared monotonicity gate failed: two SAR test curves drop by more than 0.02 as labels grow. We report the grid as descriptive.
- Checkpoints differ in objective and corpus as well as domain. We compare released checkpoints, not a causal domain effect.

Code and checkpoints



GitHub: code, evidence



Hugging Face: weights

References

Paolo et al., xView3-SAR, NeurIPS 2022. Zhou et al., Objects as Points, 2019. Steiner et al., How to Train Your ViT?, TMLR 2022. Woo et al., ConvNeXt V2, CVPR 2023. Straka & Gruber, SatDINO, 2025. Liu et al., SARMAE, CVPR 2026. Clasen et al., reBEN, IGARSS 2025.